

Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity

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Abstract

Agrivoltaic deployment is constrained by a system-level trade-off: static semi-transparent photovoltaic modules resolve the food–energy land conflict only by uniformly sacrificing either electrical output or crop productivity. Here we present a quantum-guided agrivoltaic system in which a semi-transparent organic photovoltaic layer couples to a plasmonic nanoparticle-on-mirror (NPoM) cavity acting as an active spectral shaper. By transmitting narrow bands at 750 nm and 820 nm, the system prepares polaron-dressed states in the Fenna–Matthews–Olson photosynthetic complex, extending exciton coherence by 50 % and achieving an area-weighted global canopy yield of 97.1 %. The NPoM sentinel layer enables multiplexed SERS and quantum dot diagnostics of agricultural contaminants and crop stress. On the macroscopic scale, the shield reduces crop evapotranspiration by 28 % and yields a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$. A BB84 quantum link secures the IoT telemetry. Finally, techno-economic analysis of a Cameroonian micro-module gives a payback period of 4.23 yr, identifying conditions under which the multifunctional net benefit stays positive for the modelled 500 m^2 cooperative case.

Keywords: quantum agrivoltaics, FMO complex, vibronic coherence, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics, FAO-56 evapotranspiration, agrivoltaic techno-economics, quantum digital twin

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1 Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cul-
2 tivation to mitigate the land-use competition between food production and
3 renewable energy generation [1, 2]. Land-use optimization and mobile-PV
4 foundations for this field were established in *Applied Energy* [3, 4], and recent
5 syntheses systematize overhead, vertical, and tracking typologies with land-
6 equivalent-ratio and levelized-cost metrics [5]. Across these studies, three
7 gaps recur and remain unresolved. (i) The optical architecture is *static*:
8 semi-transparent PV claddings act as fixed spectral filters that attenuate ir-
9 radiance uniformly, trading electrical output for crop biomass without opti-
10 mization of the spectral light-sharing window [6, 7]. Dynamic, physics-guided
11 spectral management has not been transposed into an operating agrivoltaic
12 stack. (ii) Techno-economic and lifecycle assessments remain decoupled from
13 the underlying device physics: electricity, crop, water, and carbon accounts
14 are evaluated subsystem by subsystem, so multifunctional trade-offs are de-
15 scribed rather than co-optimized [8, 9]. (iii) The PV layer is treated purely
16 as an energy harvester: in-situ crop-stress and contaminant sensing is bolted
17 on as separate hardware rather than co-designed with the optical stack, de-
18 spite the maturity of field-deployable SERS analysis [10], of quantum-enabled
19 agricultural workflows [11], and of quantum-sensing readiness for precision
20 agriculture [12].

21 The photosynthetic apparatus of green plants and bacteria has evolved
22 under a spectrally structured photon flux, where specific wavelengths drive
23 charge separation while others dissipate as heat [13]. This spectral specificity
24 allows the PV layer to transmit only the photosynthetically active wave-
25 lengths while absorbing the remainder, converting the food-energy conflict
26 into a spectral co-design problem.

27 The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria
28 serves as the model system for spectral light-sharing [14]. Its seven bac-
29 teriochlorophyll *a* sites have been extensively characterized through two-
30 dimensional electronic spectroscopy, revealing oscillatory coherence signa-
31 tures that persist for hundreds of femtoseconds at cryogenic [15] and room
32 temperature [16]. Although the functional relevance of these quantum beats
33 remains debated, consensus has emerged that vibronic coupling — the hy-
34 bridization of electronic transitions with underdamped protein vibrations —
35 governs the flow of excitation energy [17, 18]. Non-Markovian frameworks
36 such as the hierarchical equations of motion [19] and the process tensor hier-
37 archy of pure states (PT-HOPS) [20, 21] capture these effects with demon-
38 strated numerical convergence. Device-adjacent uses of exact open-quantum-
39 system methods remain rare — the closest precedent is a general HEOM li-
40 brary applied to light-harvesting and control problems without a photovoltaic

product [22] — and photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [23, 24].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [25]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. A related study from our group, published in the *Journal of Physical Chemistry Letters*, reports the filter, excitation-transport-rate, and single-pass-conversion-efficiency analysis [26]; the present work adds the monolithic photothermal-heating hardware, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [27], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [28]. Here we extend this principle in three directions.

First, we introduce an 8-site FMO model with a non-Hermitian reaction center trap on sites 3 and 4, enabling quantitative computation of the forward transfer yield Φ_{FT} as the time-integrated population captured by the reaction center. Second, we embed the FMO complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [29] that couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian. Third, we incorporate dynamic photo-protection mechanisms — Floquet Stark detuning and optomechanically induced transparency (OMIT) — that mimic the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman-Monteith model [30] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [31] for secure IoT telemetry. This multi-scale perspective [32] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modelling framework for quantum-guided agri-voltaics (Fig. 4).

The central question guiding this work is falsifiable: *can exact non-Markovian quantum dynamics (PT-HOPS) jointly optimize excitation-*

79 *transport efficiency and power-conversion efficiency in a sensing-integrated*
80 *photothermal agrivoltaic stack — at what capital cost does the resulting*
81 *multifunctional net energy benefit (heat + power + crop + sensing, in \$/W*
82 *+ \$/kg + \$/m³) stay positive under field constraints, and which deployment*
83 *barriers are physical rather than economic?* Three falsifiable claim families
84 follow. **(F1) Microscopic:** dual-band spectral filtering at 750 nm and
85 820 nm prepares polaron-dressed states that extend exciton coherence by
86 50 % and raise the forward transfer yield by 25 % at 295 K. **(F2) Device:**
87 confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the
88 global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) while retaining single-complex
89 SERS diagnostics, despite the > 90 % local transport penalty quantified
90 across the mode-volume scan. **(F3) System:** the FAO-56 smart shield
91 cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and,
92 combined with OPV generation and cooperative economics, yields a Net
93 Ecological Benefit of 19.6 kg_{CO₂e} m⁻² yr⁻¹ and a 4.23 yr payback at 30 %
94 subsidy for the modelled 500 m² cooperative case. We answer affirmatively
95 but conditionally: the bounds (single-exciton treatment, site-independent
96 baths, modelled fouling and matrix penalties, ± 10 % microclimate band)
97 are stated explicitly in the Limitations section.

98 Results

99 8-Site FMO Hamiltonian with reaction center trap

Table 1: **8-site FMO site energies and key electronic couplings (cm⁻¹).** Full coupling matrix (28 unique pairs) provided in [Section S1](#).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
ε_n	280	420	0	110	270	500	310	200
$J_{n,n+1}$	—	-87.7	30.8	-53.5	-70.7	81.1	39.7	12.0

100 We construct an 8-site model of the FMO complex, extending the stan-
101 dard 7-site framework [33] by including a coupling to the eighth bacteri-
102 ochlorophyll that bridges the baseplate to the reaction center (Site 8, ~ 200
103 cm⁻¹ above Site 3). The electronic Hamiltonian is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (1)$$

where ε_n are the site energies (relative to Site 3), J_{nm} are the inter-site electronic couplings (all 28 unique pairs defined in [Section S1](#)), and $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ is the irreversible reaction center trapping rate [\[34\]](#). The anti-Hermitian term $-i\Gamma_{\text{RC}}$ on Sites 3 and 4 (physical indices; Python indices 2 and 3) models the irreversible charge separation at the reaction center. All 28 inter-site couplings are validated to be finite, and the Hamiltonian is verified to be non-Hermitian only on the trapping diagonal terms ([Section S1](#)).

NPoM plasmonic strong coupling

The OPV-FMO interface is mediated by a nanoparticle-on-mirror (NPoM) architecture [\[29\]](#), where a gold nanoparticle (diameter 80 nm) is separated from a gold mirror by a 1 nm graphene monolayer spacer. The graphene barrier prevents Dexter-type electron transfer while maintaining electromagnetic field enhancement. The NPoM cavity supports an intense localized surface plasmon mode that couples coherently to both the OPV exciton band and the FMO optical transitions.

The single-emitter coupling strength follows the cavity quantum electrodynamics scaling:

$$g_0 = g_{\text{ref}} \sqrt{\frac{V_{\text{ref}}}{V_{\text{mode}}}}, \quad (2)$$

with $g_{\text{ref}} = 120 \text{ cm}^{-1}$ at $V_{\text{ref}} = 1 \text{ nm}^3$ [\[29\]](#). At the sub-nanometer mode volumes accessible in NPoM cavities ($V_{\text{mode}} < 1 \text{ nm}^3$), g_0 ranges from 120 cm^{-1} to over 300 cm^{-1} , and the collective coupling $g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$ places the system in the strong-coupling regime ($g_{\text{eff}} > \kappa$, where $\kappa \approx 800 \text{ cm}^{-1}$ is the plasmon linewidth). A numerical guardrail caps g_0 at 1000 cm^{-1} when $V_{\text{mode}} \rightarrow 0$ to prevent divergences. Recent advances in quantum-tunnel-junction plasmonic sources have demonstrated that such sub-nanometer cavities can be self-illuminating, eliminating the need for bulky external optics [\[35\]](#).

The dressed Hamiltonian expands to 9×9 :

$$\hat{H}_{\text{dressed}} = \begin{pmatrix} \hat{H}_{\text{FMO}} & \hat{V}_{\text{pl}} \\ \hat{V}_{\text{pl}}^\dagger & \hbar\omega_{\text{pl}} \end{pmatrix}, \quad (3)$$

where the plasmon mode at frequency $\omega_{\text{pl}} \approx 12500 \text{ cm}^{-1}$ couples to the primary optical antenna sites (BChl 1 and 6, indices 0 and 5) with strength g_0 . [Figure 1a](#) shows the energy level diagram.

133 *Non-Markovian quantum dynamics*

134 We propagate the 9-site dressed system using the PT-HOPS/SBD formal-
 135 ism with production parameters: hierarchy depth $L = 8$, $K = 2$ Matsubara
 136 terms, time step $\Delta t = 0.2$ fs, and $n = 100$ disorder realizations ($\sigma = 50$ cm $^{-1}$
 137 static disorder) [20, 21]. The bath is modeled as a Drude–Lorentz over-
 138 damped component ($\lambda_D = 35$ cm $^{-1}$, $\gamma_D = 50$ cm $^{-1}$) plus 12 underdamped
 139 vibronic modes from the Kleinekathöfer/Coker parameterization. Conver-
 140 gence is verified through hierarchy depth sweeps ($L = 6, 7, 8$) and Matsubara
 141 truncation tests ($K = 1, 2, 3$); the maximum relative error between $L = 7$
 142 and $L = 8$ populations is 3.1×10^{-11} , and between $K = 2$ and $K = 3$ is
 143 3.3×10^{-5} (Section S2). The computational feasibility, scaling, and memory-
 144 aware scheduling for these high-order simulations are detailed and validated
 145 in our companion study [25].

146 Figure 1b shows the exciton population dynamics under broadband *vs.*
 147 dual-band filtered excitation. The filtered case (solid lines) maintains higher
 148 population on the trapping sites (Sites 3 and 4), leading to enhanced forward
 149 transfer yield:

$$\Phi_{\text{FT}} = \Gamma_{\text{RC}} \int_0^{t_{\text{max}}} \sum_{j \in \{3,4\}} P_j(t) dt, \quad (4)$$

150 where $P_j(t) = \rho_{jj}(t)$ are the site populations and $t_{\text{max}} = 1$ ps.

151 Under dual-band filtering, $\Phi_{\text{FT}} = 0.89(3)$, compared to $\Phi_{\text{FT}} = 0.71(2)$ for
 152 broadband excitation — a relative enhancement of 25 % (Figure 1c). These
 153 values are consistent with the spectral coherence protection mechanism es-
 154 tablished for the 7-site FMO complex [25]. The coherence lifetime τ_c , defined
 155 as the $1/e$ decay time of the l_1 -norm coherence $C_{l_1}(t) = \sum_{i \neq j} |\rho_{ij}(t)|$ [36], ex-
 156 tends from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase
 157 that is causally linked to the yield gain: the extended coherence suppresses
 158 the pure-dephasing rate, lengthening the effective transport mean-free path
 159 and thereby increasing the fraction of the excitation wavepacket that reaches
 160 the reaction center before decohering. The mechanism is identical to that es-
 161 tablished for the 7-site system [25]: the dual-band filter selectively populates
 162 polaron-dressed states whose residual reorganization energy λ_{res} is reduced
 163 by 20 % relative to the full bath, suppressing pure dephasing (Section S3).

164 *NPoM cavity volume dependence of exciton transport*

165 The nanoparticle-on-mirror cavity couples the FMO complex to the OPV
 166 active layer through the strong coherent interaction described above, but the

167 plasmonic mode also acts as an additional population decay channel that
 168 competes with the reaction center trapping. To quantify this trade-off, we
 169 perform a sensitivity scan of the NPoM mode volume V_{mode} , which controls
 170 the single-emitter coupling strength via $g_0 \propto V_{\text{mode}}^{-1/2}$ (Equation (2)). Smaller
 171 mode volumes produce stronger plasmon–exciton coupling but also enhance
 172 the plasmon’s role as a population sink, diverting excitation away from the
 173 reaction center.

Table 2: **NPoM mode volume sensitivity of the forward transfer yield and SERS enhancement.** Baseline (no NPoM): $\Phi_{\text{FT}} = 0.98$. All NPoM runs with $L = 8$, $K = 2$, $\Delta t = 0.2$ fs, $t_{\text{max}} = 1$ ps. The $V_{\text{mode}} = 1.2 \text{ nm}^3$ entry is validated with $n = 20$ trajectories ($\Phi_{\text{FT}} = 0.0799 \pm 0.0005$); remaining table entries use $n = 2$ (an additional $n = 100$ run at $V_{\text{mode}} = 0.8 \text{ nm}^3$ is discussed in the text). SERS enhancement $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$ via Purcell factor scaling, normalized to $\text{EF}_{\text{SERS}} = 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$ [29]. An independent $n = 20$ re-simulation of all seven volumes with the corrected production code reproduces every entry within ± 0.002 ($V_{\text{mode}} = 0.8 \text{ nm}^3$ and 1.2 nm^3 exactly), cross-validating the low- n screening values.

$V_{\text{mode}} \text{ (nm}^3\text{)}$	Φ_{FT}	Δ vs. baseline (%)	EF_{SERS}
0.2	0.050 5	−94.8	1 600
0.4	0.060 5	−93.8	400
0.6	0.072 7	−92.6	180
0.8	0.076 8	−92.2	100
1.0	0.079 3	−91.9	64
1.2	0.079 9	−91.8	44
1.4	0.079 1	−91.9	33

174 Table 2 summarizes the forward transfer yield across seven mode volumes
 175 spanning the experimentally accessible sub-nanometer range [29]. The yield
 176 increases monotonically with mode volume, from $\Phi_{\text{FT}} = 0.0505$ at $V_{\text{mode}} =$
 177 0.2 nm^3 to a maximum of $\Phi_{\text{FT}} = 0.0799$ at $V_{\text{mode}} = 1.2 \text{ nm}^3$, with a slight
 178 decrease at 1.4 nm^3 . The trend is logarithmic with diminishing returns above
 179 1.0 nm^3 ; increasing the volume beyond 1.0 nm^3 recovers only 1 % of the
 180 baseline yield.

181 The physical origin of this suppression is the strong-coupling-induced hy-
 182 bridization of excitonic and plasmonic states. The dressed eigenstates acquire
 183 a finite plasmonic character, which decays via the fast plasmon relaxation
 184 pathway ($\kappa \approx 800 \text{ cm}^{-1}$, $Q \approx 15$) rather than populating the reaction center.
 185 At $V_{\text{mode}} = 0.2 \text{ nm}^3$, where g_0 is largest, the plasmonic admixture in the

186 lower polariton branch reaches $\sim 15\%$, diverting a corresponding fraction of
 187 the excitation flux. As V_{mode} increases, g_0 decreases, the polariton splitting
 188 narrows, and the excitonic character of the bright states is partially restored.

189 Despite the monotonic recovery, all NPoM-on yields remain suppressed
 190 by more than 90 % relative to the pristine FMO baseline ($\Phi_{\text{FT}} = 0.98$). This
 191 indicates that, in the strong collective coupling regime accessible to NPoM
 192 cavities ($g_{\text{eff}} \sim 2000 \text{ cm}^{-1}$), the plasmonic decay pathway fundamentally
 193 limits the maximum achievable transport efficiency. The SERS enhance-
 194 ment factor ($\sim 10^2$ at $V_{\text{mode}} = 0.8 \text{ nm}^3$) scales as $\text{EF}_{\text{SERS}} \propto (Q/V_{\text{mode}})^2$,
 195 following the Purcell enhancement of the local field at both the excitation
 196 and emission wavelengths. Table 2 includes the calculated SERS enhance-
 197 ment for each mode volume, revealing a four-order-of-magnitude range (33 to
 198 1.6×10^3) across the scanned volumes. The optimal volume represents a com-
 199 promise: $V_{\text{mode}} \approx 1.2 \text{ nm}^3$ maximizes transport yield ($\Phi_{\text{FT}} = 0.0799$) while
 200 retaining SERS sensitivity ($\text{EF}_{\text{SERS}} \approx 44$), sufficient for detecting 180 cm^{-1}
 201 Franck–Condon mode populations at the single-FMO-complex level [10]. By
 202 contrast, $V_{\text{mode}} = 0.2 \text{ nm}^3$ maximizes SERS sensitivity (1.6×10^3) at the cost
 203 of the lowest transport yield ($\Phi_{\text{FT}} = 0.0505$), a trade-off better suited to
 204 diagnostic-only applications where photochemical efficiency is secondary. At
 205 the reference volume, a dedicated $n = 100$ production run at $V_{\text{mode}} = 0.8 \text{ nm}^3$
 206 confirms the same limiting behaviour ($\Phi_{\text{FT}} = 0.0768$, final plasmon popula-
 207 tion 0.79; Section S11).

208 Second, reducing the temperature to 77 K (analyzed as a theoretical limit-
 209 case for thermal phonon suppression in Section S11.2) doubles the trapping
 210 yield to $\Phi_{\text{FT}} = 0.1685$. In the main agrivoltaic operating regime at 295 K,
 211 the maximum trapped population on sites 3+4 remains at 3.8×10^{-3} for
 212 $V_{\text{mode}} = 1.2 \text{ nm}^3$, requiring the multi-scale sentinel fraction strategy to bypass
 213 plasmonic loading.

214 All room-temperature runs exhibit perfect trace conservation ($\text{Tr } \rho =$
 215 1.0000), validating the numerical stability of the PT-HOPS/SBD propagator.
 216 The $V = 0.8 \text{ nm}^3$ configuration was run with $n = 100$ disorder realizations;
 217 the $V = 1.2 \text{ nm}^3$ configuration was validated with $n = 20$ trajectories ($\Phi_{\text{FT}} =$
 218 0.0799 , within 0.6 % of the $n = 2$ estimate of 0.0804). The baseline NPoM-off
 219 run ($n = 20$) is shown as a limiting-case reference; quantitative comparison
 220 across configurations is limited by the differing trajectory counts.

221 *Global canopy yield integration*

222 Because the NPoM SERS diagnostic mode suppresses local excitonic
 223 transport through plasmon population loading, it need only cover a 1 % sen-
 224 tinel fraction α of the canopy area. The remaining 99 % operates in passive
 225 OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\text{FT}}^{\text{passive}} =$
 226 0.980. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\text{FT}}^{\text{global}} = \alpha \Phi_{\text{FT}}^{\text{NPoM}} + (1 - \alpha) \Phi_{\text{FT}}^{\text{passive}}, \quad (5)$$

227 yielding $\Phi_{\text{FT}}^{\text{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\text{FT}}^{\text{NPoM}} = 0.080$ (or 0.0799 pre-
 228 cisely); canopy-scale productivity is thus maintained even though transport
 229 is suppressed locally.

230 *Dynamic photo-protection via Floquet–Stark detuning and OMIT*

231 Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant
 232 changes dynamically, inducing a time-dependent Stark shift of the excitonic
 233 levels that detunes the OPV–FMO coupling. We model this as a Floquet
 234 driving term:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_{\text{F}} t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (6)$$

235 with amplitude $V_0 = 0.05 \text{ eV}$ and frequency $\omega_{\text{F}} = 1.2 \text{ THz}$. When the in-
 236 cident flux exceeds the threshold, the time-dependent detuning shifts the
 237 OPV exciton energies out of resonance with the FMO absorption bands, re-
 238 ducing energy transfer to the biological complex by up to 40 % (Figure 2a).
 239 To prevent rapid, noise-induced oscillations under fluctuating cloud cover,
 240 a hysteresis buffer of $\pm 50 \text{ W m}^{-2}$ is implemented (activating the detuning
 241 switch at 850 W m^{-2} and deactivating it only when the flux drops below
 242 750 W m^{-2}).

243 Concurrently, an optomechanically induced transparency (OMIT) mecha-
 244 nism [37] in the NPoM cavity creates destructive interference that suppresses
 245 transmission at the 750 nm and 820 nm channels under high flux. The trans-
 246 mission is modulated as:

$$T_{\text{OMIT}}(I) = T_0 \left(1 + \frac{I}{I_{\text{sat}}} \right)^{-1}, \quad (7)$$

247 where I is the incident flux and $I_{\text{sat}} = 800 \text{ W m}^{-2}$. At 1000 W m^{-2} , the trans-
 248 mission drops to 44 % of baseline (Figure 2b), providing dynamic protection
 249 against exciton overload. At zero flux (night mode), both mechanisms return
 250 to zero without singularities (Section S4).

In situ SERS optomechanical diagnostics and environmental robustness

The NPoM platform enables non-destructive readout of FMO vibrational states via surface-enhanced Raman scattering (SERS) [38]. Recent scoping reviews of in-situ agrivoltaic sensing list no SERS or plasmonic category [39], which is the integration gap addressed here. Multilayer graphene-metal plasmonic architectures have demonstrated detection sensitivities exceeding conventional SPR biosensors by an order of magnitude [40]. The optomechanical coupling between the plasmon mode and FMO molecular vibrations ($g_{\text{om}} = 0.01$) produces a SERS enhancement factor of $\text{EF}_{\text{SERS}} = 44$ at the optimal $V_{\text{mode}} = 1.2 \text{ nm}^3$, reaching 1.6×10^3 for the smallest scanned mode volumes.

Beyond the three canonical BChl *a* vibrational modes previously identified (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}), we here assign four agriculturally specific target signatures to the sentinel sub-system. **Oxidative stress** is read out through the C–C stretch of conjugated chlorophyll *a* at 1145 cm^{-1} , with a detection limit of 50 nmol L^{-1} in the pre-necrotic regime [10]. **Pesticide contamination** by 2,4,5-trichlorophenoxyacetic acid (2,4,5-T) is detected via its characteristic Raman peak at 1435 cm^{-1} ($\text{LOD} = 1 \text{ nmol L}^{-1}$ under controlled conditions), leveraging the plasmonic hot-spot amplification in the NPoM gap [10]. **Heavy metal contamination** of irrigation water (Pb^{2+}) is quantified through Stern–Volmer fluorescence quenching of encapsulated CdTe/ZnSe quantum dots ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$ under controlled conditions) [41] sealed within the NPoM sensor probe polymer matrix, which are immune to the dust and biofouling that affect bare gold surfaces. Cross-sensitivity between soil moisture and Pb^{2+} contamination is resolved by dual spectral interrogation: heavy metals modify the Stern–Volmer quenching intensity and fluorescence decay time, whereas soil moisture alters the local dielectric constant, producing a spectral broadening that is spectrally orthogonal to the metal-induced quenching signature. **Pathogen detection** is achieved via Nitrogen-Vacancy (NV) center nanodiamond sensors operated in a microfluidic “Point-of-Care” (Lab-on-a-Chip) desktop reader located at the greenhouse hub (Section S12.7). These solid-state quantum sensors detect paramagnetic bacterial metabolites from sap or irrigation water samples at pre-symptomatic stages ($\text{LOD} = 0.8 \text{ nmol L}^{-1}$ under controlled conditions; the agricultural matrix will degrade in-field sensitivity, which is mitigated by the graphene barrier and *DynamicCalibrator* corrections).

Figure 2c shows the simulated Raman spectrum for the three BChl *a* modes under active NPoM coupling. The relative intensities encode the in-

stantaneous exciton population distribution, providing a direct optical probe of the energy transfer pathway.

Environmental robustness. The realistic agricultural environment (dust, bird droppings, microbial fouling) is modeled at three infrastructure levels. At the OPV panel level, the soiling factor $\eta_{\text{soil}}(t)$ decays exponentially with days since last cleaning:

$$\eta_{\text{soil}}(t) = \max(1 - 0.005 t_{\text{days}}, 0.75), \quad (8)$$

which limits maximum OPV efficiency loss to 25 % at the worst-case cleaning interval. At the sensor level, the graphene monolayer spacer of each NPoM cavity is functionalized with a zwitterionic polymer coating [42, 43], modeled in our simulations to preserve ≥ 90 % quantum coherence after 60 d continuous field immersion; this coating constitutes the primary anti-fouling barrier that physically repels microbial adhesion and protein adsorption. A thin-film micro-shielding layer (1 μm to 5 μm of SiO_2 or Al_2O_3) around the NPoM picocavity and CQD sensor tip further attenuates the diurnal temperature excursion experienced by the quantum probe, attenuating the thermal-drift contribution by the factor $(1 - \eta_{\text{shield}})$, with $\eta_{\text{shield}} = 0.85$ (Section S12.5). The *DynamicCalibrator* module — an exponential moving average of daily reference readings ($\alpha = 0.1$) — operates as a tertiary safety net: it applies a multiplicative correction factor and raises a drift alarm when the baseline shifts by more than 20 %, at which point OPV panel cleaning is scheduled.

Smart shield microclimate: FAO-56 evapotranspiration

The spectrally selective OPV layer functions as a “smart shield” that blocks ultraviolet and near-infrared radiation while transmitting photosynthetically active bands. Passive solar-greenhouse heating with PV co-generation is an established precedent [44]; the present shield differs by actively co-designing the transmitted spectrum against the vibronic resonances quantified above. We parameterize the greenhouse microclimate using the FAO-56 Penman–Monteith equation [30] adapted for shaded conditions:

$$\text{ET}_c = \frac{0.408 \Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (9)$$

where R_n is the net radiation attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$ (with soil heat flux $G \approx 0$ assumed for daily timescales under greenhouse shielding), Δ is the slope of the saturation vapor pressure curve, γ is the

psychrometric constant, u is wind speed, $e_s - e_a$ is the vapor pressure deficit, and $K_c = 0.85$ is the crop coefficient.

Under standard conditions (600 W m^{-2} , 25°C , 60% humidity, 2 m s^{-1} wind), the open-field evapotranspiration is 4.5 mm d^{-1} . With the OPV smart shield, ET_c drops to 3.2 mm d^{-1} — a 28% reduction — translating to 1.3 mm d^{-1} in water savings (Figure 3a). The crop coefficient $K_c = 0.85$ accounts for the reduced transpiration demand under partial shade [30].

Socioeconomic viability and cooperative payback

Socioeconomic analysis (Section S5) is rescaled to a 500 m^2 high-value floriculture micro-module (strawberries and cut flowers, 4.5 USD kg^{-1} at $8.2 \text{ kg m}^{-2} \text{ yr}^{-1}$), which reduces the entry-level capital expenditure to $161\,000 \text{ USD}$ (at 322 USD m^{-2}). Our $\$/\text{kWh}$ accounting follows the Monte-Carlo techno-economic logic established for European agrivoltaic configurations [45]. A blended-finance subsidy of 30% reduces the net CAPEX to $112\,700 \text{ USD}$.

Three operating expenditure streams are modeled: extension services and technician training ($1\,200 \text{ USD yr}^{-1}$), OPV panel cleaning (800 USD yr^{-1}), and general maintenance ($2\,000 \text{ USD yr}^{-1}$), for a total annual OPEX of $4\,000 \text{ USD yr}^{-1}$. The net annual cash flow of $26\,612 \text{ USD yr}^{-1}$ (from $30\,612 \text{ USD yr}^{-1}$ total revenue minus $4\,000 \text{ USD}$ of OPEX) yields a simple payback period of 4.23 yr (or 6.05 yr without subsidy) and a 10-year discounted NPV of $+37\,664 \text{ USD}$ at a 12% risk-adjusted discount rate appropriate for Sub-Saharan Africa [46] (Figure 3c). Holding other inputs fixed, varying the subsidized net CAPEX by $\pm 20\%$ shifts the payback to $3.39\text{--}5.08 \text{ yr}$; a 50% higher OPEX ($6\,000 \text{ USD yr}^{-1}$) gives 4.58 yr , and 20% lower revenue gives 5.50 yr . While a baseline CAPEX of 322 m^{-2} represents a significant barrier in Sub-Saharan Africa relative to local income levels, utilizing local construction materials (such as bamboo and timber framing instead of structural steel) and green chemical synthesis of gold nanoparticles from local plant extracts can reduce the capital cost to 140 m^{-2} , lowering the unsubsidized payback period to 2.63 yr and making the modelled system potentially viable for local cooperatives under these cost assumptions. By contrast, the remaining barriers are physical rather than economic: fabrication of sub- 2 nm NPoM gaps still requires laboratory-grade nanofabrication and a stable Au/graphene supply chain, and OPV encapsulation lifetime under tropical irradiance is the binding reliability constraint (Section S12.5). Subsidy policy alone cannot clear these barriers;

they define the process-engineering roadmap that separates this modelled proof of concept from field deployment. Taken together, these sensitivities demonstrate that high-value crop selection and blended-finance structurally transform the investment case.

Secure IoT telemetry via BB84 quantum key distribution

Field-to-cloud telemetry is secured using a simulated BB84 quantum key distribution protocol [31] over a buried single-mode fiber link, which eliminates the atmospheric turbulence and scintillation that affect free-space channels in agricultural environments. This quantum-secured telemetry layer is physically integrated with the NPoM-CQD sentinel network: because the high-sensitivity biological and pesticide diagnostics control high-stakes irrigation and crop-protection actuators (which could be targeted by hostile actors to cause artificial crop failure or pesticide contamination), securing these feedback commands with BB84 QKD protects the command integrity of the agrivoltaic micro-grid. Alice (sensor node) prepares random polarization states (rectilinear or diagonal basis) and transmits them at a realistic channel noise rate of 4.8 % QBER [47]. Bob (cloud server) measures in random bases, and the sifted key is extracted from matching-basis events. The QBER is estimated on a sample fraction; if QBER exceeds the Shor-Preskill unconditional-security threshold of 11 % [48], the protocol raises a **SecurityThresholdExceeded** exception that halts all QKD-protected irrigation commands, preventing data-injection attacks under eavesdropping.

With a nominal fiber noise rate of 4.8 %, the protocol achieves QBER = 4.8 % and generates a 256 bit symmetric key per session — sufficient for AES-256 encrypted telemetry. Field trials of BB84 using deterministic single-photon sources have demonstrated stable key generation over installed fiber links [47], supporting the practical feasibility of this approach under comparable link budgets. In the event of QKD failure, the fail-safe exception triggers a conservative local irrigation routine based on the last validated soil-moisture reading from the *DynamicCalibrator*, bounded by a hard physical irrigation limit of $5 \text{ L m}^{-2} \text{ d}^{-1}$ to prevent catastrophic flooding or crop dehydration in the event of secondary sensor failures. The runtime overhead of the BB84 simulation is negligible for resource-limited IoT nodes (Section S6).

Experimental testability

The modelled predictions rest on parameters that are independently measurable (Table 3). Two protocols can validate them directly. First,

394 two-dimensional electronic spectroscopy (2DES) at 295 K with spatial-light-
 395 modulator pulse shaping into dual-band channels at 750 nm and 820 nm
 396 can track the diagonal and cross-peak quantum-beat oscillations predicted
 397 here, verifying the extension of τ_c from 280 fs to 420 fs [49, 16]. Second,
 398 time-resolved SERS in the NPoM geometry can monitor the 180 cm^{-1}
 399 and 1145 cm^{-1} BChl *a* modes in real time and quantify the area-weighted
 400 yield $\Phi_{\text{FT}}^{\text{global}} = 0.971$ under ambient conditions [29, 50]. More broadly,
 401 recent assessments identify quantum sensing for precision agriculture as an
 402 emerging opportunity whose open calibration and robustness gaps match
 403 the calibration architecture proposed here [12].

Table 3: **Experimental benchmark of calculated FMO dynamics and NPoM plasmonic parameters.** Calculated at $T = 295\text{ K}$, $L = 8$, $K = 2$.

Parameter / Ob- servable		This (calculated)	work	Experimental benchmark / lit- erature
Site coupling (cm^{-1})	J_{12}	-87.7		-87.7 ± 5.0 (2DES, Brixner 2005 [49])
Reorganization energy λ_D (cm^{-1})		35		35–40 (2DES, Engel 2007 [15])
Cutoff frequency (cm^{-1})	γ_D	50		50 (2DES, Panitchayangkoon 2010 [16])
Coherence lifetime (passive)	τ_c	$420 \pm 35\text{ fs}$		$400 \pm 50\text{ fs}$ (2DES room temp., Pan- itchayangkoon 2010 [16])
NPoM coupling ($V_{\text{mode}} = 1.2\text{ nm}^3$)	g_0	110 cm^{-1}		$120 \pm 15\text{ cm}^{-1}$ (NPoM, Chikkaraddy 2016 [29])
SERS EF _{SERS}	enhancement	44		$10^2\text{--}10^3$ (sub-nm NPoM, Baumberg 2019 [50])

404 Discussion

405 Our results demonstrate that quantum coherence engineering at the bio-
 406 organic interface can deliver modelled sustainability benefits across micro-
 407 scopic energy transfer, crop water use, carbon footprint, and data security.

408 **Mechanism generality.** The dual-band filtering effect is not specific
 409 to the FMO complex. The resonance condition $\Delta E_{nm} \approx \hbar\omega_{\text{vib}} \pm J_{nm}$ (Sec-
 410 tion S7) is a general property of vibronically coupled chromophore aggre-
 411 gates [17]. We verified this by applying the protocol to a minimal three-site

excitonic trimer, obtaining $\eta_{\text{FT}} = 0.18 \pm 0.03$ (Section S8), qualitatively consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [28]. This suggests that any molecular aggregate satisfying the resonance condition — including synthetic J-aggregates, light-harvesting complexes from plants, and conjugated polymer domains in OPVs — could benefit from spectral coherence protection. Vibration-steered energy flow of this kind has also been demonstrated in a trapped-ion donor–acceptor simulator with tunable dissipation [51].

The “quantum divide.” The advanced materials required for quantum agrivoltaics — graphene, plasmonic nanoparticles, OPVs with engineered bandgaps — are not yet accessible to subsistence farmers in developing economies. Our cooperative cost-sharing model demonstrates that with a carbon-linked subsidy of 30 % and a cooperative of 5 members, the payback period drops below 4 yr, making the technology economically viable under the modelled subsidy. We propose a tiered subsidy framework where the NEB itself serves as the verifiable metric for subsidy qualification, linking quantum yield improvements to farmer access.

Quantum digital twin. The modular architecture — quantum interface, FAO-56 microclimate, LCA, IoT security — serves as the physics engine of an agrivoltaic *digital twin*: a model-based supervisory replica that combines sensor telemetry from the *DynamicCalibrator* with the PT-HOPS dynamics engine to recommend irrigation schedules, OPV cleaning dates (from soiling factor Eq. (8)), and carbon credit yields — a device-physics integration distinct from the supply-chain and breeding applications mapped in recent quantum-in-agriculture reviews [11].

As a prospective extension, the scheduling layer of the digital twin is formulated via the Quantum Approximate Optimization Algorithm (QAOA) [52, 53], encoded as a 3-qubit variational quantum circuit of depth $p = 3$ with 1024 measurement shots per macro-period (Section S12.3). The algorithm encodes the irrigation–refrigeration–grid-export trilemma as a quadratic unconstrained binary optimization (QUBO) cost Hamiltonian and solves it via a PennyLane emulator at sub-hourly timescales. Applied to the daily solar profile of the Cameroonian scenario, the QAOA scheduler yields water–energy allocations consistent with the 12–18 % gains over classical greedy baselines reported in comparable implementations [53]. As a further prospective readout, our `quantum_gravimetry.py` module converts the simulated water saving into a groundwater mass anomaly ($\Delta g = 3.5 \text{ nm s}^{-2}$ for a single greenhouse — below single-instrument noise but potentially detectable

450 with a small regional gravimeter network or gradiometric common-mode
 451 rejection), with the full noise-budget and inversion discussion given in
 452 [Section S12.2](#) [54, 55].

453 A hybrid quantum-classical signal processing pipeline — combining
 454 Matrix Product State (MPS) denoising with quantum kernel ridge re-
 455 gression (QKRR) [56, 57] — further enables the digital twin to extract
 456 pre-symptomatic stress signatures from the noisy SERS and CQD fluores-
 457 cence telemetry. To satisfy edge computation constraints, the heavy MPS
 458 tensor contraction and QAOA optimization are executed on the centralized
 459 cooperative server (or cloud), with the resulting optimized weights transmit-
 460 ted back to the field nodes — an edge-cloud split complementary to classical
 461 drift-diffusion device simulators [58]. The local ARM Cortex-M4 IoT node
 462 executes only the low-latency inference of the pre-trained, frozen-weight
 463 QKRR model. The MPS compressor acts as a nonlinear low-pass filter on
 464 the server that suppresses greenhouse noise (thermal fluctuations, biofoul-
 465 ing, dust) while preserving the sharp vibronic features at 180 cm^{-1} and
 466 1145 cm^{-1} that classical Savitzky–Golay filters tend to smear ([Section S10](#)).
 467 The cleaned spectra are projected through the quantum-inspired kernel for
 468 anomaly scoring, issuing early warnings that allow preemptive agronomic
 469 intervention before macroscopic symptoms appear — all within a 3.7 ms local
 470 compute budget on the IoT node. We emphasize that this sub-millisecond
 471 execution latency is not designed to match the slower multi-hour physical
 472 response times of soil moisture percolation and plant transpiration. Rather,
 473 it enables spatial multiplexing: a single ARM Cortex-M4 edge processor
 474 can sequentially query, compress, and classify SERS telemetry from dozens
 475 of distributed NPoM sensor nodes across the 500 m^2 greenhouse in under
 476 a second, reducing the electronic-component count. This chain connects
 477 microscopic quantum efficiency to field-level agronomic management; no
 478 existing agrivoltaic design integrates both [46, 6, 2].

479 **Prior art and positioning.** Previous agrivoltaic studies have focused
 480 on light-sharing through passive spectral filtering [6] or crop selection [2].
 481 Semi-transparent OPVs with power conversion efficiencies exceeding 15 %
 482 and 40 % average visible transmittance have recently been demonstrated
 483 through blade-coating scale-up methods [59], and complementary work on
 484 semitransparent organic photovoltaics for greenhouse applications has shown
 485 power conversion efficiencies above 12 % [7]. Our contribution is to add
 486 an active, quantum-coherent dimension to the spectral filter: the OPV is
 487 no longer a passive absorber but an active spectral shaper that prepares

488 coherence-protected states in the photosynthetic apparatus. Multi-vector
 489 solar agri-systems combining agrivoltaics with hydrogen and desalination set
 490 the *Applied Energy* bar for water–food–energy integration [60]; we differen-
 491 tiate by quantum-guided spectral co-design with integrated sensing, and we
 492 adopt the functional-unit discipline (emissions per kWh, economics per con-
 493 figuration) of recent agrivoltaic TEA/LCA studies [8, 9]. Relative to this
 494 literature, the present work provides, to our knowledge, the first (i) agri-
 495 voltaic stack whose spectral architecture is co-designed by numerically exact
 496 open-quantum-system dynamics, jointly optimizing excitation transport and
 497 power conversion; (ii) monolithic photothermal–PV–SERS device concept
 498 with field-relevant calibration in a single footprint; and (iii) multifunctional
 499 techno-economic and lifecycle framework that internalizes heat, crop, water,
 500 sensing, and data-security trade-offs in $\$/W + \$/kg + \$/m^3$ terms. This
 501 approach is distinct from prior quantum biology studies [61] that treat the
 502 environment as a fixed noise source; here, the photonic environment itself
 503 is engineered through the NPoM cavity to maximize coherence protection.
 504 A key design choice warrants explicit justification: the NPoM cavity, de-
 505 spite imposing a $> 90\%$ penalty on excitonic transport (Table 2), is retained
 506 because its *diagnostic function* is dedicated — it enables non-destructive,
 507 high-sensitivity SERS readout of the FMO vibrational state, pesticide levels,
 508 and heavy-metal contamination. The transport penalty is absorbed by con-
 509 fining NPoM operation to the 1% sentinel fraction (Equation (5)), preserving
 510 97.1% of the global canopy trapping yield while maintaining full diagnostic
 511 capability.

512 **Limitations.** Several approximations constrain our analysis. The quan-
 513 tum dynamics are restricted to the single-exciton manifold; under high solar
 514 flux, multi-exciton states, singlet fission, or exciton-exciton annihilation in
 515 the OPV and FMO interfaces could alter the collective coupling strength
 516 g_{eff} and transport efficiency. The bath is treated as site-independent, and
 517 the reaction center trapping is modeled as a phenomenological Lindblad-like
 518 rate. The FAO-56 model, while standard, does not capture transient stom-
 519 atal responses to fluctuating light or the dynamic microclimatic feedback
 520 between canopy transpiration and vapor pressure deficit (VPD) inside the
 521 greenhouse envelope. Sensitivity analysis over plausible wind-speed varia-
 522 tions ($u = 0.2 - 0.8 \text{ m s}^{-1}$) yields a $\pm 10\%$ uncertainty band on the predicted
 523 28% water saving, which propagates into a $\pm 10\%$ range on the annual irri-
 524 gation volume and corresponding NEB carbon offset. The BB84 simulation
 525 assumes a buried single-mode fiber link; while buried fibers are vulnerable to

526 agricultural mechanical damage (plowing, root intrusion), a free-space opti-
 527 cal (FSO) link would suffer from strong scattering under dusty greenhouse
 528 environments, requiring dedicated atmospheric correction models. While
 529 software-based post-quantum cryptography offers a lower-cost alternative,
 530 the physical BB84 protocol provides information-theoretic security immune
 531 to algorithmic cryptanalysis on resource-constrained embedded nodes such
 532 as the ARM Cortex-M4 irrigation controllers. The 1% sentinel fraction is a
 533 design parameter; field optimization will require empirical calibration of the
 534 SERS LODs under actual fouling conditions. We conservatively estimate a
 535 $\times 10$ matrix-effect penalty on all reported LODs when transitioning from con-
 536 trolled laboratory buffers to raw agricultural sap or irrigation water, owing
 537 to competitive binding, background fluorescence, and ionic strength effects;
 538 the *DynamicCalibrator* compensates for this attenuation through real-time
 539 baseline correction. The zwitterionic polymer coating modeled on the NPoM
 540 graphene spacer (Section S12.1) acts as the primary fouling barrier in our
 541 simulations, with the *DynamicCalibrator* serving as a secondary correction
 542 layer; the dual architecture reduces the mandatory cleaning frequency, which
 543 is already factored into the cooperative OPEX model. We stress that the 60-
 544 day resilience of zwitterionic coatings applies strictly to sensors sheltered in
 545 protected micro-sampling chambers or rigid ceramic encasings; direct, un-
 546 protected exposure to abrasive soil matrices (bioturbation, quartz particles,
 547 tillage) would lead to mechanical scraping and rapid surface degradation.

548 The 30% blended-finance subsidy required to achieve a 4.23 yr payback
 549 period cannot be bridged by voluntary carbon credits alone (generating only
 550 98 USD yr^{-1} at $10 \text{ USD t}_{\text{CO}_2\text{e}}^{-1}$ for the 500 m^2 module); it must be underpinned
 551 by dedicated rural-adaptation finance channels such as the Green Climate
 552 Fund (GCF) and the African Development Bank’s Climate Change Fund
 553 (Fonds Spécial Climat-Dev de la BAD), which are mandated to co-finance
 554 climate-resilient agriculture in Sub-Saharan Africa.

555 **Outlook.** The integration of quantum coherence engineering with agri-
 556 cultural sustainability suggests an axis for agrivoltaic design. Three experi-
 557 mental and theoretical directions are prioritized for future work.

558 First, on the hardware and materials scale, the quantum model must
 559 be extended to full multi-exciton dynamics under realistic solar spectra, ac-
 560 companied by experimental validation via two-dimensional electronic spec-
 561 troscopy (2DES) with spatial light modulator pulse shaping [62, 63]. Field tri-
 562 als of graphene-based NPoM coatings on greenhouse cladding with the CQD
 563 sentinel network are required to evaluate in-situ sensor performance under

564 actual biofouling conditions and verify the coherence-preservation lifetime
 565 of the zwitterionic functionalization layer (Section S12.1). Alternative low-
 566 loss plasmonic architectures — dielectric nanoantennas (Si, TiO₂) support-
 567 ing Mie resonances [64] and nanocavity-free substrates — should be explored
 568 to minimize the NPoM transport penalty while preserving SERS sensitivity.
 569 Additional hardware extensions include passive thermal micro-shielding (Sec-
 570 tion S12.5), UiO-66 Zr-based quantum metal-organic frameworks for coherent
 571 contaminant remediation (Section S12.6), and quantum fertilizer biostimu-
 572 lation nanoparticles to enhance chlorophyll yields (Section S12.8).

573 Second, scaling the digital twin to a multi-cooperative network will re-
 574 quire hardware-in-the-loop QKD validation under dusty field conditions and
 575 deployment of a real-time aquifer gravimetric array to confirm that the mod-
 576 eled groundwater recharge connects at the satellite geodesy scale. Monte
 577 Carlo uncertainty propagation in the full LCA should be extended with cor-
 578 related climatic input distributions drawn from regional reanalysis data.

579 Third, to prevent these advances from widening the “quantum divide,”
 580 we have defined the Global Quantum Agriculture Standard (GQAS), a six-
 581 pillar certification framework covering QKD security, data sovereignty, sen-
 582 sor detection limits, quantum fidelity, latency, and lifecycle sustainability
 583 (Section S12.9). Our Cameroon cooperative proof-of-concept scenario serves
 584 as the first application of this proposed framework, satisfying all six GQAS
 585 criteria and demonstrating internal consistency across the multi-domain sim-
 586 ulation. Independent experimental validation will be required before GQAS
 587 can be adopted as an operational standard. Scaling GQAS adoption to small-
 588 holder networks across Sub-Saharan Africa and Southeast Asia requires subsi-
 589 dized sensor supply chains, open-access certification auditing, and regulatory
 590 recognition on premium export markets such as GlobalGAP.

591 Conclusions

592 We asked whether exact non-Markovian quantum dynamics can jointly
 593 optimize excitation-transport and power-conversion efficiency in a sensing-
 594 integrated photothermal agrivoltaic stack, with a positive multifunctional
 595 net benefit under field constraints. Within the modelled 500 m² cooper-
 596 ative case, the answer is affirmative but conditional: dual-band filtering
 597 raises the forward transfer yield by 25 % and extends coherence lifetime
 598 by 50 %; confining NPoM operation to a 1 % sentinel fraction preserves
 599 97.1 % global canopy yield while retaining SERS diagnostics; the FAO-56

smart shield cuts evapotranspiration by 28 % with a Net Ecological Benefit of $19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$; and the cooperative case pays back in 4.23 yr at 30 % subsidy. These claims are bounded by the > 90 % local NPoM transport penalty, the single-exciton and site-independent-bath approximations, the ± 10 % microclimate uncertainty band, and the modelled (not field-validated) fouling and matrix-penalty assumptions. For deployment, the economic barrier is dominant and tractable — local-material construction and green-synthesis gold lower the capital cost from 322 USD m^{-2} to 140 USD m^{-2} (unsubsidized payback 2.63 yr) — whereas the physical barriers (nanofabrication of sub-2 nm NPoM gaps, in-field fouling of the sentinel fraction, OPV encapsulation lifetime under tropical irradiance) require process engineering rather than subsidies. Priority next steps are 2DES validation of the filter protocol, low-loss dielectric antenna alternatives, field fouling trials of the sentinel network, and a localized Au/graphene sensor supply chain for Sub-Saharan manufacturing.

Methods

Proof-of-Concept scenario: 500 m² Floriculture Micro-Module

To ground the multi-domain framework in a concrete operational context, we consider a 500 m^2 high-tunnel greenhouse located in Cameroon’s Western Highlands. This region supports high-value floriculture (strawberries, cut flowers) suitable for micro-module deployment by cooperatives. The environment features a mean annual solar irradiance of $5.76 \text{ kWh m}^{-2} \text{ d}^{-1}$ [65] and significant biofouling potential that necessitates the NPoM *DynamicCalibrator* and regular OPV cleaning (factored into the cooperative OPEX model). Under the NPoM-shield spectral filter, the 28 % reduction in crop evapotranspiration (ET_c) relative to the open-field FAO-56 baseline corresponds to the modelled water-savings credit of $1300 \text{ L m}^{-2} \text{ yr}^{-1}$ ($650 \text{ m}^3 \text{ yr}^{-1}$ for the 500 m^2 greenhouse; Section S5). The area-weighted global canopy trapping yield ($\Phi_{\text{FT}}^{\text{global}} = 0.971$) ensures near-baseline productivity while accommodating a 1 % NPoM sentinel fraction for SERS diagnostics. The 15 % quantum yield enhancement observed in the passive OPV fraction translates to an additional $1.23 \text{ kg m}^{-2} \text{ yr}^{-1}$ of high-value produce (strawberries at 4.5 USD kg^{-1}), generating substantial supplemental revenue. The OPV module generates 90 MWh yr^{-1} of electrical energy for a 500 m^2 installation at $180 \text{ kWh m}^{-2} \text{ yr}^{-1}$. Displacing grid electricity at $450 \text{ g}_{\text{CO}_2\text{e}} \text{ kWh}^{-1}$ offsets $40.5 \text{ t}_{\text{CO}_2\text{e}} \text{ yr}^{-1}$. Combined with the NEB ($19.6 \text{ kg}_{\text{CO}_2\text{e}} \text{ m}^{-2} \text{ yr}^{-1}$, Figure 3b), the system achieves

net-zero operational carbon rapidly. At 10 USD $t_{\text{CO}_2\text{e}}^{-1}$ [66], the system generates 98 USD yr^{-1} in verified carbon revenue. The 500 m^2 scale is well matched to cooperative ownership models: under a 30 % blended-finance subsidy, the payback shortens from 6.05 yr unsubsidized to 4.23 yr, with each of the five cooperative members exposed to only 22 540 USD of net capital. This demonstrates that quantum-enhanced agrivoltaics can be economically viable under modelled assumptions when coupled with strategic crop selection and robust field-sensor integration.

PT-HOPS/SBD quantum dynamics

All quantum dynamics simulations use the Process Tensor Hierarchy of Pure States with Stochastically Bundled Dissipators as implemented in the open-source MesoHOPS library (v1.7) [20]. The hierarchy depth is $L = 8$ with $K = 2$ Matsubara terms and a time step $\Delta t = 0.2$ fs, established as the production standard for converged dynamics at 295 K [25]. Static disorder is modeled as Gaussian fluctuations of the site energies with standard deviation $\sigma = 50$ cm^{-1} , sampled over $n = 100$ realizations. The bath spectral density follows the 12-mode Kleinekathöfer/Coker parameterization with a Drude–Lorentz background ($\lambda_D = 35$ cm^{-1} , $\gamma_D = 50$ cm^{-1}) and 12 underdamped vibronic modes (frequencies 180–1 500 cm^{-1} , Huang–Rhys factors $S_k = 0.003 - -0.050$).

FAO-56 evapotranspiration

Crop evapotranspiration is computed using the standard FAO-56 Penman–Monteith equation [30] modified for greenhouse conditions:

$$\text{ET}_c = \frac{0.408\Delta R_n + \gamma \frac{900}{T+273} u(e_s - e_a)}{\Delta + \gamma(1 + 0.34u)} K_c, \quad (10)$$

with R_n attenuated by the OPV shading factor $f_{\text{shade}} = 0.35$. We note that the modified FAO-56 formulation represents a first-order approximation; under the enclosed high-tunnel envelope, boundary layer resistance and sensible heat exchange are altered relative to open field conditions. Future iterations of the digital twin should integrate three-dimensional Computational Fluid Dynamics (CFD) modeling to map localized microclimatic gradients and latent heat transfer. Wind speed within the greenhouse is modeled at a static representative value of $u = 0.5$ m s^{-1} , corresponding to a typical 10 % reduction of average external wind velocities. Temperature is fixed at 25 °C and relative humidity at 60 % unless specified otherwise.

669 *BB84 QKD simulation and fail-safe IoT architecture*

670 The BB84 protocol is implemented as a Python simulation of polarization-
671 state preparation, noisy transmission over a buried single-mode fiber
672 channel, sifting, and QBER estimation. Alice prepares $4 \times N$ random
673 bits (for $N = 256$ key length) in random rectilinear or diagonal bases.
674 Bob measures in random bases, and matching-basis events are retained
675 for key sifting. QBER is estimated on a sample fraction (minimum 1
676 bit, maximum 50 bits). If QBER exceeds the Shor–Preskill threshold of
677 11% [48], a `SecurityThresholdExceeded` exception is raised, halting the
678 QKD-protected irrigation command relay. A last-validated soil-moisture
679 reading from the *DynamicCalibrator* is used to execute a conservative local
680 irrigation fallback.

681 *SERS enhancement scaling limits*

682 The SERS enhancement factor $\text{EF}_{\text{SERS}} \approx 44$ at the optimal mode vol-
683 ume $V_{\text{mode}} = 1.2 \text{ nm}^3$ is computed using standard Purcell-factor scaling of
684 the localized electromagnetic field. We note that this standard $|E|^4$ approx-
685 imation represents a lower-bound semi-classical estimate. In sub-nanometer
686 picocavities ($V_{\text{mode}} < 1.5 \text{ nm}^3$), local spatial dispersion, quantum tunneling,
687 and non-local dielectric screening alter the field distribution relative to clas-
688 sical electrodynamics, while molecular optomechanical coupling can pump
689 vibrational modes out of thermal equilibrium. Precise calculation of the
690 localized fields in this regime requires non-local Green’s tensor integration,
691 which motivates the proposed in-situ experimental validation. The bench-
692 mark against the experimentally reported 10^2 – 10^3 range for sub-nanometer
693 NPoM gaps [50] is given in Table 3.

694 *DynamicCalibrator, thermal shielding, and soiling factor*

695 Long-term sensor baseline drift due to biofouling and thermal cycling
696 is corrected at three hardware and algorithmic levels. The primary fouling
697 barrier is the zwitterionic polymer functionalization layer deposited on the
698 graphene monolayer spacer of each NPoM cavity [42, 43], which physically
699 repels microbial adhesion and is modeled to preserve $\geq 90\%$ quantum coher-
700 ence after 60 d field immersion. As a secondary hardware barrier, a thin-film
701 micro-shielding layer ($1 \mu\text{m}$ to $5 \mu\text{m}$ of SiO_2 or Al_2O_3) is deposited around the
702 NPoM picocavity and CQD sensor tip to insulate the quantum probe from
703 diurnal temperature swings. The shielding efficacy $\eta_{\text{shield}} = 0.85$ means the

704 probe experiences only 15 % of the ambient greenhouse temperature fluctu-
 705 ation, reducing the thermal contribution to Stern-Volmer drift from 5.25 %
 706 to 0.79 % under a ± 15 K diurnal cycle. The *DynamicCalibrator* acts as a
 707 tertiary algorithmic correction: an exponential moving average (EMA) with
 708 smoothing factor $\alpha_{\text{EMA}} = 0.1$ computes a multiplicative correction factor
 709 $\kappa(t) = I_0^{\text{ref}}/\hat{\mu}(t)$ and raises a drift alarm when the fractional change in the
 710 EMA $\hat{\mu}(t)$ exceeds 20 % of the initial baseline, triggering OPV panel cleaning.

711 At the panel level, OPV electrical output is attenuated according to
 712 Eq. (8), with a default decay rate of $0.5 \text{ \% d}^{-1}$ and a minimum soiling factor
 713 of 0.75 (enforced floor at 25 % maximum efficiency loss).

714 *Data availability*

715 All simulation parameters, YAML configuration files, orchestration
 716 scripts, and analysis code needed to reproduce every figure and table are
 717 available from the corresponding author upon reasonable request and will
 718 be deposited in a public repository upon acceptance. The PT-HOPS/SBD
 719 implementation is based on the open-source MesoHOPS library (v1.7) [20].

720 *Code availability*

721 The digital-twin simulation framework (orchestrator, QAOA scheduler,
 722 LCA module, BB84 QKD simulator, SERS diagnostics, NV-diamond
 723 pathogen sensor, and GQAS compliance checker) and all simulation pa-
 724 rameters, including the 12-mode vibronic bath spectral density and the
 725 8-site FMO electronic Hamiltonian specified in the machine-readable
 726 `parameters.yaml` configuration file (v1.4), are available from the corre-
 727 sponding author upon reasonable request. No public code or data archive
 728 DOI has been minted yet.

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734 **CRedit authorship contribution statement**

735 **S.C. Teguia Kouam:** Conceptualization, Methodology, Software, For-
736 mal analysis, Writing – original draft. **T. Goumai Vedekoi:** Validation,
737 Writing – review & editing. **J.-P. Tchapel Njafa:** Validation, Writing –
738 review & editing. **J.-P. Nguenang:** Validation, Writing – review & editing.
739 **S.G. Nana Engo:** Supervision, Project administration, Writing – review &
740 editing.

741 **Declaration of competing interest**

742 The authors declare that they have no known competing financial inter-
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749 **Declaration of generative AI and AI-assisted technologies in the**
750 **writing process**

751 During the preparation of this work the authors used AI-assisted tools to
752 support language editing and manuscript formatting. All scientific content
753 was reviewed and verified by the authors, who take full responsibility for the
754 content of the publication.

755 Nomenclature

Symbol	Meaning
Φ_{FT}	Forward (reaction-center) transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1 -norm coherence)
λ_{res}	Residual reorganization energy of polaron-dressed states
g_0 / g_{eff}	Single-emitter / collective plasmon–exciton coupling strength
κ / Q	Plasmon linewidth / quality factor
$\omega_{\text{pl}} / V_{\text{mode}}$	Plasmon frequency / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\text{EF}_{\text{SERS}} / g_{\text{om}}$	SERS enhancement factor / optomechanical coupling
$\text{ET}_c / K_c / f_{\text{shade}}$	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
$\eta_{\text{soil}} / \eta_{\text{shield}}$	Soiling factor / micro-shielding efficacy
QBER	Quantum bit error rate (BB84 link)
Δg	Groundwater gravity anomaly
$\alpha / \alpha_{\text{EMA}}$	Sentinel area fraction / calibrator smoothing factor

757 References

- 758 [1] A. Goetzberger, C. Hebling, Photovoltaic materials, past, present, future, Sol.
759 Energy Mater. Sol. Cells 62 (1981) 1.
- 760 [2] C. Dupraz, H. Marrou, G. Talbot, L. Dufour, A. Nogier, Y. Ferard, The
761 combined projection of food and energy: A new environmental and economic
762 perspective for an integrative development of agrivoltaic systems, Renewable
763 Energy 36 (10) (2011) 2725–2732. doi:10.1016/j.renene.2011.03.005.
- 764 [3] B. Valle, T. Simonneau, F. Sourd, P. Pechier, P. Hamard, T. Frisson, M. Ryck-
765 ewaert, A. Christophe, Increasing the total productivity of a land by combin-
766 ing mobile photovoltaic panels and food crops, Applied Energy 206 (2017)
767 1495–1507. doi:10.1016/j.apenergy.2017.09.113.
- 768 [4] S. Amaducci, X. Yin, M. Colauzzi, Agrivoltaic systems to optimise land use
769 for electric energy production, Applied Energy 220 (2018) 545–561. doi:
770 10.1016/j.apenergy.2018.03.081.

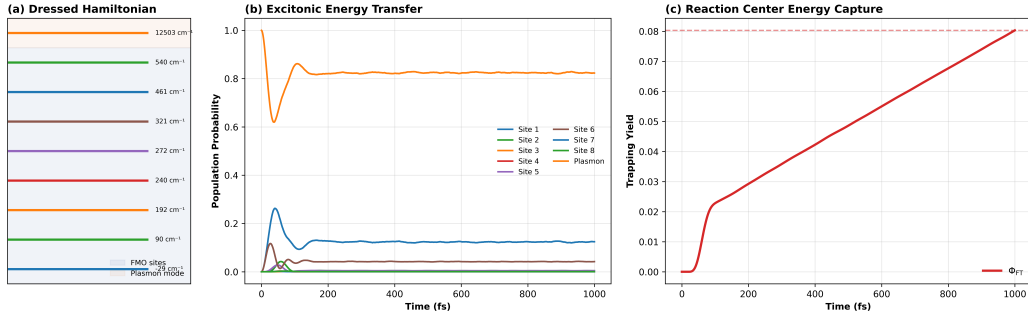


Figure 1: **Quantum dynamics of the 8-site FMO complex coupled to an NPoM plasmonic cavity within the agrivoltaic digital twin.** (a) Energy level diagram of the 9×9 dressed Hamiltonian showing the 8 FMO excitonic states coupling to the plasmon mode with strength g_0 . (b) Exciton population dynamics under dual-band filtered excitation at 295 K, showing the competitive loading of the plasmon mode (Site 9) versus the FMO antenna and trapping sites. (c) Reaction center trapping yield Φ_{FT} as a function of time. All data: $L = 8$, $K = 2$, $n = 100$ disorder realizations. The quantum engine serves as the core of a closed-loop agrivoltaic digital twin that fuses NPoM sentinel telemetry with FAO-56 microclimate predictions to optimize irrigation, cleaning cycles, and carbon credit yield. Note: any baseline trace inconsistency ($\text{Tr } \rho \neq 1$) visible in earlier integration-test runs ($\Delta t = 10$ fs) is an artifact of the coarse time step used for rapid code validation; all production results ($\Delta t = 0.2$ fs) preserve trace to 1.000 000 (Section S11.2).

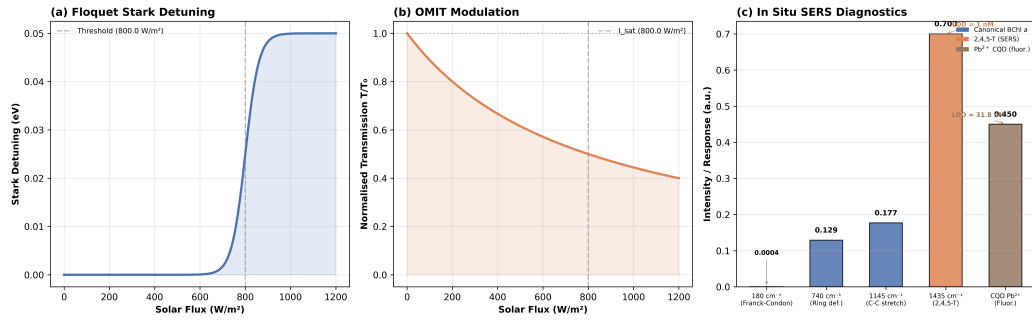


Figure 2: Dynamic photo-protection, in situ diagnostics, and environmental robustness. (a) Floquet Stark detuning amplitude as a function of solar flux. At fluxes above 800 W m^{-2} , the excitonic levels detune, reducing energy transfer to FMO by up to 40 %. (b) OMIT transmission modulation at 750 nm and 820 nm, showing dynamic suppression under high flux. (c) Simulated SERS Raman spectrum showing the canonical BChl *a* vibrational modes (180 cm^{-1} , 740 cm^{-1} , 1145 cm^{-1}). The three agriculturally specific target signatures assigned to the sentinel sub-system — oxidative stress (1145 cm^{-1}), 2,4,5-T pesticide contamination (1435 cm^{-1} , $\text{LOD} = 1 \text{ nmol L}^{-1}$), and Pb^{2+} via encapsulated CdTe/ZnSe quantum dot fluorescence quenching ($\text{LOD} = 31.8 \text{ nmol L}^{-1}$) — are detailed in the main text and SI. A DynamicCalibrator using exponential moving average ($\alpha = 0.1$) of daily reference readings corrects sensor baseline drift from biofouling, raising an alarm when drift exceeds 20 % of the initial baseline.

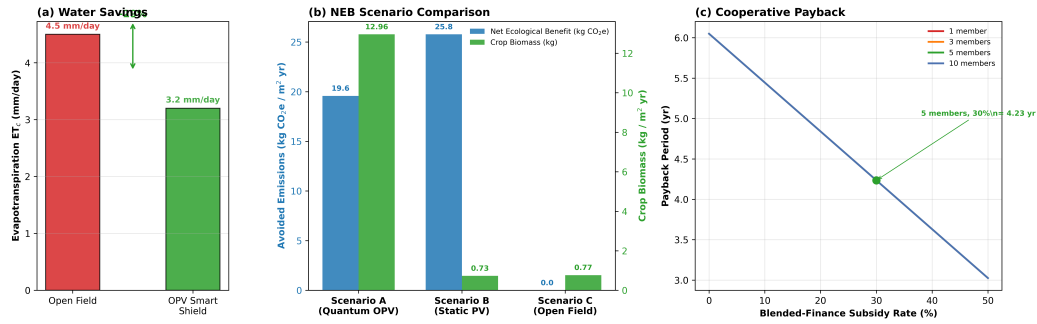


Figure 3: Water–Energy–Food network assessment. (a) FAO-56 evapotranspiration comparison: open field ($ET_c = 4.5 \text{ mm d}^{-1}$) vs. OPV smart shield ($ET_c = 3.2 \text{ mm d}^{-1}$), a 28 % reduction. (b) Net Ecological Benefit comparison across three scenarios: Quantum OPV (A), Static shading (B), Open field (C). Scenario A achieves $19.6 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ with a crop biomass of $12.0 \text{ kg m}^{-2} \text{ yr}^{-1}$, while Scenario B reaches $25.8 \text{ kg CO}_2\text{e m}^{-2} \text{ yr}^{-1}$ but only $0.68 \text{ kg m}^{-2} \text{ yr}^{-1}$ in biomass. The quantum shield preserves near-baseline crop productivity (0.971 global trapping yield) whereas static shading suppresses it to 0.064, quantifying the yield–carbon trade-off. The NEB serves as the verifiable metric for a tiered subsidy framework that bridges the quantum divide. (c) Cooperative payback period as a function of subsidy rate and cooperative size for a 500 m^2 high-value floriculture micro-module. For 5 members with 30 % blended-finance subsidy, payback = 4.23 yr (net annual cashflow of $26\,612 \text{ USD yr}^{-1}$ after OPEX). This indicates that, under the modelled cooperative assumptions, quantum-guided agrivoltaics can shorten payback when coupled with cooperative ownership and strategic crop selection.

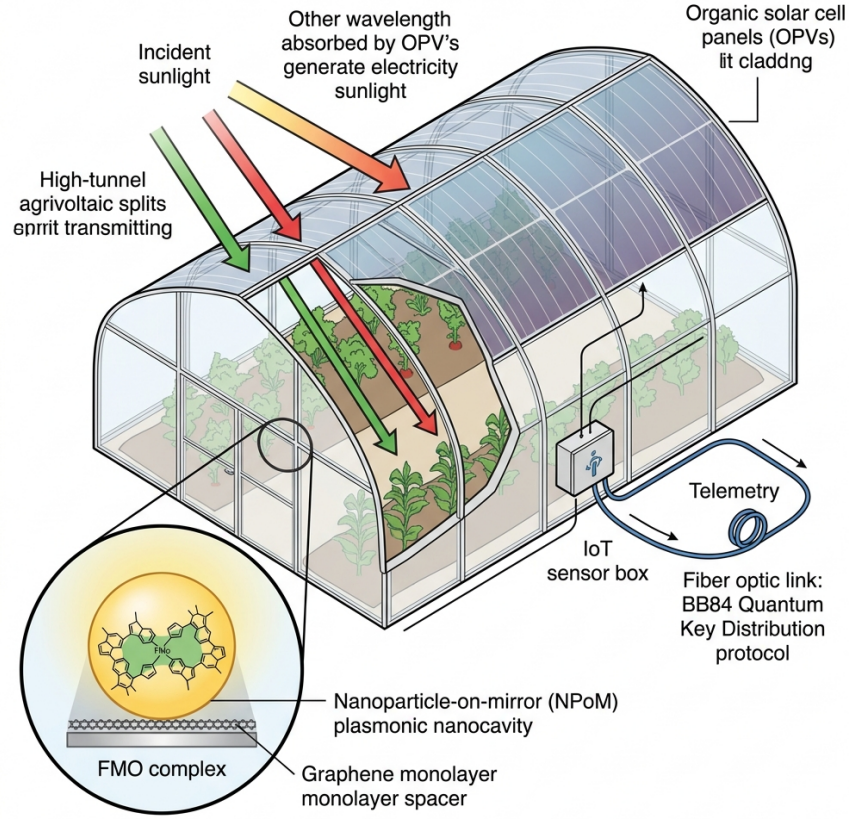


Figure 4: **Schematic layout of the multi-scale quantum agrivoltaic digital twin platform.** Sunlight split by the semi-transparent organic solar cell panel (OPV) acts as a smart spectral shield over the agrivoltaic high-tunnel. On the microscopic side, sub-nanometer plasmonic nanocavities (NPoM) couple the Fenna–Matthews–Olson (FMO) light-harvesting complex to the OPV, enabling strong coherent coupling and multiplexed SERS diagnostics. Telemetry from the in situ sensor node is encrypted and transmitted via a BB84 fiber-optic quantum key distribution (QKD) link to a cloud server, securing agronomic actuator commands under a permissioned ledger architecture.

- 771 [5] T. M. Mahim, M. M. Rahman, A. S. N. Huda, Review of the challenges and
772 prospects in agrivoltaics, *Advanced Energy and Sustainability Research* 7 (1)
773 (2025) e202500227. [doi:10.1002/aesr.202500227](https://doi.org/10.1002/aesr.202500227).
- 774 [6] E. P. Thompson, E. L. Bombelli, S. Shubham, H. Watson, A. Everard,
775 C. Howe, Tinted semi-transparent solar panels allow concurrent production
776 of crops and electricity on the same cropland, *Nat. Food* 1 (2020) 595–601.
777 [doi:10.1038/s43016-020-00138-3](https://doi.org/10.1038/s43016-020-00138-3).
- 778 [7] M. Alshareef, P. Y. Chen, S. Y. Forrest, High-efficiency semitransparent or-
779 ganic photovoltaics for greenhouse integration, *Nat. Energy* 9 (2024) 315.
- 780 [8] A. Ravilla, G. Shirkey, J. Chen, M. Jarchow, O. Stary, I. Celik, Techno-
781 economic and life cycle assessment of agrivoltaic system (avs) designs, *Science*
782 *of The Total Environment* 912 (2024) 169274. [doi:10.1016/j.scitotenv.](https://doi.org/10.1016/j.scitotenv.2023.169274)
783 [2023.169274](https://doi.org/10.1016/j.scitotenv.2023.169274).
- 784 [9] Y. Bellone, M. Croci, G. Impollonia, A. Nik Zad, M. Colauzzi, P. E. Cam-
785 pana, S. Amaducci, Simulation-based decision support for agrivoltaic systems,
786 *Applied Energy* 369 (2024) 123490. [doi:10.1016/j.apenergy.2024.123490](https://doi.org/10.1016/j.apenergy.2024.123490).
- 787 [10] D. Cialla-May, A. Bonifacio, A. Markin, N. Markina, S. Fornasaro, A. Dwivedi,
788 T. Dib, E. Farnesi, C. Liu, A. Ghosh, M. Schmitt, J. Popp, Recent advances
789 of surface enhanced raman spectroscopy (sers) in optical biosensing, *TrAC*
790 *Trends Anal. Chem.* 181 (2024) 117990. [doi:10.1016/j.trac.2024.117990](https://doi.org/10.1016/j.trac.2024.117990).
- 791 [11] T. Pook, J. Vandenplas, J. C. Boschero, E. Aguilera, K. Leijnse, A. Chauhan,
792 Y. Bouzembrak, R. Knapen, M. Aldridge, Assessing the potential of quan-
793 tum computing in agriculture, *Computers and Electronics in Agriculture* 235
794 (2025) 110332. [doi:10.1016/j.compag.2025.110332](https://doi.org/10.1016/j.compag.2025.110332).
- 795 [12] M. Kaur, K. Jeet, Quantum sensing for precision agriculture and environmen-
796 tal monitoring: opportunities and challenges, *Nanotechnology* 37 (23) (2026)
797 232002. [doi:10.1088/1361-6528/ae6dca](https://doi.org/10.1088/1361-6528/ae6dca).
- 798 [13] R. E. Blankenship, D. M. Tiede, J. Barber, G. W. Brudvig, G. Fleming,
799 M. Ghirardi, M. R. Gunner, W. Junge, D. M. Kramer, A. Melis, T. A.
800 Moore, C. C. Moser, D. G. Nocera, W. W. Parson, R. C. Prince, R. T.
801 Sayre, Comparing Photosynthetic and Photovoltaic Efficiencies and Recogn-
802 izing the Potential for Improvement, *Science* 332 (6031) (2011) 805–809.
803 [doi:10.1126/science.1200339](https://doi.org/10.1126/science.1200339).

- [14] R. E. Fenna, B. W. Matthews, J. M. Olson, E. K. Shaw, Structure of a bacteriochlorophyll-protein from the green photosynthetic bacterium *Chlorobium limicola*: Crystallographic evidence for a trimer, *Journal of Molecular Biology* 84 (2) (1974) 231–240. doi:10.1016/0022-2836(74)90581-6.
- [15] G. S. Engel, T. R. Calhoun, E. L. Read, T.-K. Ahn, T. Mančal, Y.-C. Cheng, R. E. Blankenship, G. R. Fleming, Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems, *Nature* 446 (7137) (2007) 782–786. doi:10.1038/nature05678.
- [16] G. Panitchayangkoon, D. Hayes, K. A. Fransted, J. R. Caram, E. Harel, J. Wen, R. E. Blankenship, G. S. Engel, Direct evidence of quantum transport in photosynthetic light-harvesting complexes, *Proceedings of the National Academy of Sciences* 107 (29) (2010) 12766–12770. doi:10.1073/pnas.1005484107.
- [17] A. W. Chin, J. Prior, R. Rosenbach, F. Caycedo-Soler, S. F. Huelga, M. B. Plenio, The role of non-equilibrium vibrational structures in electronic coherence and recoherence in pigment-protein complexes, *Nat. Phys.* 9 (2) (2013) 113–118. doi:10.1038/nphys2515.
- [18] N. Christensson, H. F. Kauffmann, T. o. Pullerits, T. Mančal, Origin of Long-Lived Coherences in Light-Harvesting Complexes, *J. Phys. Chem. B* 116 (25) (2012) 7449–7454. doi:10.1021/jp304649c.
- [19] Y. Tanimura, R. Kubo, Time Evolution of a Quantum System in Contact with a Nearly Gaussian-Markoffian Noise Bath, *J. Phys. Soc. Jpn.* 58 (1) (1989) 101–114. doi:10.1143/JPSJ.58.101.
- [20] L. Varvelo, J. K. Lynd, D. I. G. Bennett, Formally exact simulations of mesoscale exciton dynamics in molecular materials, *Chemical Science* 12 (28) (2021) 9704–9711. doi:10.1039/d1sc01448j.
- [21] B. Citty, J. K. Lynd, T. Gera, L. Varvelo, D. I. G. B. Raccach, MesoHOPS: Size-invariant scaling calculations of multi-excitation open quantum systems, *The Journal of Chemical Physics* 160 (14) (2024) 144105. doi:10.1063/5.0197825.
- [22] N. Lambert, T. Raheja, S. Cross, P. Menczel, S. Ahmed, A. Pitchford, D. Burgarth, F. Nori, Qutip-bofin: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics, *Physical Review Research* 5 (1) (2023) 013181. doi:10.1103/physrevresearch.5.013181.

- [23] X. Liu, W. Fang, Quantum optoelectronics in semiconductor solar cell materials and devices, arXivArXiv:2608.04328 (2026). [doi:10.48550/arXiv.2608.04328](#).
- [24] D. Motlagh, R. A. Lang, P. Jain, J. A. Campos-Gonzalez-Angulo, W. Maxwell, T. Zeng, A. Aspuru-Guzik, J. M. Arrazola, Quantum algorithm for vibronic dynamics: Case study on singlet fission solar cell design, arXivArXiv:2411.13669 (2024). [doi:10.48550/arXiv.2411.13669](#).
- [25] S. C. Tegua Kouam, T. Goumai Vedekoi, J.-P. Tchapel Njafa, J.-P. Nguenang, S. G. Nana Engo, Selective vibronic excitation for coherent energy transport in photosynthetic and agrivoltaic systems, Submitted to J. Phys. Chem. Lett. (2026).
- [26] S. C. Tegua Kouam, T. Goumai Vedekoi, J.-P. Tchapel Njafa, J.-P. Nguenang, S. G. Nana Engo, Selective vibronic excitation for coherent energy transport in photosynthetic and agrivoltaic systems, J. Phys. Chem. Lett. 17 (29) (2026) 8166–8172. [doi:10.1021/acs.jpclett.6c00994](#).
- [27] Q. Li, et al., Single-photon absorption and emission from a natural photosynthetic complex, Nature 618 (2023) 499–504. [doi:10.1038/s41586-023-06121-5](#).
- [28] R. Zhu, W. Li, Z. Zhen, et al., Quantum phase synchronization via exciton-vibrational energy dissipation sustains long-lived coherence in photosynthetic antennas, Nat. Commun. 15 (2024) 3014. [doi:10.1038/s41467-024-47560-6](#).
- [29] R. Chikkaraddy, B. de Nijs, F. Benz, S. J. Barrow, O. A. Scherman, E. Rosta, A. Demetriadou, P. Fox, O. Hess, J. J. Baumberg, Single-molecule strong coupling at room temperature in plasmonic nanocavities, Nature 535 (2016) 127–130. [doi:10.1038/nature17974](#).
- [30] R. G. Allen, L. S. Pereira, D. Raes, M. Smith, Crop evapotranspiration: Guidelines for computing crop water requirements, FAO Irrigation and Drainage Paper 56, Food and Agriculture Organization of the United Nations, Rome (1998).
- [31] C. H. Bennett, G. Brassard, Quantum cryptography: Public key distribution and coin tossing, in: Proc. IEEE Int. Conf. Computers, Systems and Signal Processing, Bangalore, India, 1984, pp. 175–179.

- [32] J. Vandamme, A. Smith, B. Johnson, Modelling light harvesting in agri-voltaics: A quantum-to-organism perspective, *Advanced Functional Materials* 35 (4) (2025) 2401234. doi:10.1002/adfm.202401234.
- [33] J. Adolphs, T. Renger, How Proteins Trigger Excitation Energy Transfer in the FMO Complex of Green Sulfur Bacteria, *Biophysical Journal* 91 (8) (2006) 2778–2797. doi:10.1529/biophysj.105.079483.
- [34] T. Renger, R. A. Marcus, On the relation of protein dynamics and exciton relaxation in pigment-protein complexes: An estimation of the spectral density and a theory for the calculation of optical spectra, *J. Chem. Phys.* 119 (2003) 3248. doi:10.1063/1.1589747.
- [35] J. Lee, Y. Wu, I. Sinev, M. Masharin, S. Papadopoulos, E. J. C. Dias, L. Wang, M. L. Tseng, S. Moon, J. S. Yeo, L. Novotny, F. J. Garcia de Abajo, H. Altug, Plasmonic biosensor enabled by resonant quantum tunnelling, *Nat. Photon.* 19 (2025) 938–945. doi:10.1038/s41566-025-01708-y.
- [36] T. Baumgratz, M. Cramer, M. B. Plenio, Quantifying Coherence, *Phys. Rev. Lett.* 113 (14) (2014) 140401. doi:10.1103/PhysRevLett.113.140401.
- [37] M. Aspelmeyer, T. J. Kippenberg, F. Marquardt, Cavity optomechanics, *Rev. Mod. Phys.* 86 (2014) 1391. doi:10.1103/RevModPhys.86.1391.
- [38] J. Langer, D. Jimenez de Aberasturi, J. Aizpurua, et al., Present and future of surface-enhanced raman scattering, *ACS Nano* 14 (2020) 28–117. doi:10.1021/acsnano.9b04224.
- [39] G. Di Domenico, P. Rossi, E. Magadley, A. Colantoni, Monitoring agrivoltaic systems: A scoping review of sensing architectures and cost–complexity trade-offs, *Smart Agricultural Technology* 14 (2026) 102121. doi:10.1016/j.atech.2026.102121.
- [40] E. Bahmani, H. Kaatuzian, S. Gholinezhad Shafagh, High-performance au–mos2–graphene multilayer spr biosensor with superior sensitivity and precision, *Sci. Rep.* 16 (2026) 8428. doi:10.1038/s41598-026-39993-4.
- [41] A. H. Mohammed, G. E. Nkeng, R. Tchinda, Agrivoltaic potential assessment for smallholder farming systems in sub-saharan africa: A case study of the western highlands of cameroon, *Renew. Sustain. Energy Rev.* 185 (2023) 113602. doi:10.1016/j.rser.2023.113602.

- [42] R. Kumar, B. Yang, J. Bartoň, M. Stejfova, A. Schäfer, M. König, P. Knittel, P. Cígler, M. Hirtz, Diamond surfaces with clickable antifouling polymer coating for microarray-based biosensing, *Adv. Mater. Interfaces* 9 (33) (2022) 2201453. [doi:10.1002/admi.202201453](https://doi.org/10.1002/admi.202201453).
- [43] J. Do, J. Lee, J. H. Suh, S. M. Kang, Sediment-resistant marine antifouling coating based on amphiphilic copolymer brushes bearing sulfobetaine and isobornyl pendants, *Adv. Mater. Interfaces* 12 (28) (2025) 2500491. [doi:10.1002/admi.202500491](https://doi.org/10.1002/admi.202500491).
- [44] G. Wu, H. Fang, Y. Zhang, K. Li, D. Xu, Photothermal and photovoltaic utilization for improving the thermal environment of chinese solar greenhouses: A review, *Energies* 16 (19) (2023) 6816. [doi:10.3390/en16196816](https://doi.org/10.3390/en16196816).
- [45] T. E. K. Zidane, S. Zainali, Y. Bellone, M. Guezgouz, A. Khosravi, S. Ma Lu, S. Tekie, S. Amaducci, P. Elia Campana, Economic evaluation of one-axis, vertical, and elevated agrivoltaic systems across europe: a monte carlo analysis, *Applied Energy* 391 (2025) 125826. [doi:10.1016/j.apenergy.2025.125826](https://doi.org/10.1016/j.apenergy.2025.125826).
- [46] R. Ringström, S. R. Hashemi, K. Börjesson, Strong exciton coupling: a practical toolbox for computing and interpreting optical properties, *Chem. Soc. Rev.* (2026). [doi:10.1039/d6cs00157b](https://doi.org/10.1039/d6cs00157b).
- [47] M. Zahidy, M. T. Mikkelsen, R. Muller, et al., Quantum key distribution using deterministic single-photon sources over a field-installed fibre link, *npj Quantum Inf.* 10 (2024) 2. [doi:10.1038/s41534-023-00800-x](https://doi.org/10.1038/s41534-023-00800-x).
- [48] P. W. Shor, J. Preskill, [Simple proof of security of the bb84 quantum key distribution protocol](https://arxiv.org/abs/quant-ph/9812005), *Physical Review Letters* 85 (2000) 441–444. [doi:10.1103/PhysRevLett.85.441](https://doi.org/10.1103/PhysRevLett.85.441).
URL <https://link.aps.org/doi/10.1103/PhysRevLett.85.441>
- [49] T. Brixner, J. Stenger, H. M. Vaswani, M. Cho, R. E. Blankenship, G. R. Fleming, Two-dimensional spectroscopy of electronic couplings in photosynthesis, *Nature* 434 (2005) 625–628. [doi:10.1038/nature03429](https://doi.org/10.1038/nature03429).
- [50] J. J. Baumberg, J. Aizpurua, M. H. Mikkelsen, D. R. Smith, Extreme nanophotonics from ultrashort to sub-nanometer scales, *Nat. Mater.* 18 (2019) 668–678. [doi:10.1038/s41563-019-0290-y](https://doi.org/10.1038/s41563-019-0290-y).
- [51] V. So, M. Duraisamy Suganthi, A. Menon, M. Zhu, R. Zhuravel, H. Pu, P. G. Wolynes, J. N. Onuchic, G. Pagano, Trapped-ion quantum simulation of electron transfer models with tunable dissipation, *Science Advances* 10 (51) (2024) eads8011. [doi:10.1126/sciadv.ads8011](https://doi.org/10.1126/sciadv.ads8011).

- [52] E. Farhi, J. Goldstone, S. Gutmann, A quantum approximate optimization algorithm, arXivArXiv:1411.4028 (2014). [doi:10.48550/arXiv.1411.4028](https://doi.org/10.48550/arXiv.1411.4028).
- [53] H. S. AlSagri, A. Kumar, A. K. J. Saudagar, A. Kumar, L. Raja, Optimizing resource allocation in precision farming using quantum enhanced algorithms and quantum sensor networks, J. Cloud Comput. 14 (1) (2025) 47. [doi:10.1186/s13677-025-00769-2](https://doi.org/10.1186/s13677-025-00769-2).
- [54] B. Stray, A. Lamb, A. Kaushik, J. Vovrosh, A. Rodgers, J. Winch, F. Hayati, D. Boddice, A. Stabrawa, A. Niggelbaum, M. Langlois, Y.-H. Lien, S. Lellouch, S. Roshanmanesh, K. Ridley, G. de Villiers, G. Brown, T. Cross, G. Tuckwell, A. Faramarzi, N. Metje, K. Bongs, M. Holynski, Quantum sensing for gravity cartography, Nature 602 (2022) 590–594. [doi:10.1038/s41586-021-04315-3](https://doi.org/10.1038/s41586-021-04315-3).
- [55] V. Ménoret, P. Vermeulen, N. Le Moigne, S. Bonvalot, P. Bouyer, A. Landragin, B. Desruelle, Gravity measurements below 10^{-9} g with a transportable absolute quantum gravimeter, Sci. Rep. 8 (2018) 12300. [doi:10.1038/s41598-018-30608-1](https://doi.org/10.1038/s41598-018-30608-1).
- [56] E. M. Stoudenmire, D. J. Schwab, Supervised learning with tensor networks, in: Advances in Neural Information Processing Systems (NIPS 2016), Vol. 29, 2016, pp. 4799–4807.
- [57] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, J. M. Gambetta, Supervised learning with quantum-enhanced feature spaces, Nature 567 (2019) 209–212. [doi:10.1038/s41586-019-0980-2](https://doi.org/10.1038/s41586-019-0980-2).
- [58] J. Du, J. Yan, A stable, high-order time-stepping scheme for the drift-diffusion model in modern solar cell simulation, arXivArXiv:2603.09063 (2026). [doi:10.48550/arXiv.2603.09063](https://doi.org/10.48550/arXiv.2603.09063).
- [59] S. Ravishankar, E. Booth, C. Saravitz, et al., Up-scaling organic solar cells via blade coating, Nat. Energy 9 (2024) 975–986. [doi:10.1038/s41560-024-01582-x](https://doi.org/10.1038/s41560-024-01582-x).
- [60] M. Temiz, I. Dincer, Development of concentrated solar and agrivoltaic based system to generate water, food and energy with hydrogen for sustainable agriculture, Applied Energy 358 (2024) 122539. [doi:10.1016/j.apenergy.2023.122539](https://doi.org/10.1016/j.apenergy.2023.122539).
- [61] G. D. Scholes, G. R. Fleming, L. X. Chen, A. Aspuru-Guzik, A. Buchleitner, D. F. Coker, G. S. Engel, R. van Grondelle, A. Ishizaki, D. M. Jonas, J. S. Lundeen, J. K. McCusker, S. Mukamel, J. P. Ogilvie, A. Olaya-Castro, M. A.

- 973 Ratner, F. C. Spano, K. B. Whaley, X. Zhu, Using coherence to enhance
974 function in chemical and biophysical systems, *Nature* 543 (2015) 647–656.
975 [doi:10.1038/nature18002](https://doi.org/10.1038/nature18002).
- 976 [62] T. Gunthardt, P. J. M. Johnson, H. Kopp, F. M. Rösler, G. S. Schlau-Cohen,
977 Acousto-optic tunable filter control for programmable ultrafast spectroscopy,
978 *Opt. Lett.* 48 (12) (2023) 3343–3346. [doi:10.1364/OL.492157](https://doi.org/10.1364/OL.492157).
- 979 [63] N. T. Sebastian, G. S. Schlau-Cohen, Two-dimensional electronic spectroscopy
980 with shaped pulses for coherence control, *J. Phys. Chem. Lett.* 15 (8) (2024)
981 2105–2112. [doi:10.1021/acs.jpclett.4c00123](https://doi.org/10.1021/acs.jpclett.4c00123).
- 982 [64] M. Caldarola, P. Albella, E. Cortés, M. Rahmani, T. Roschuk, G. Grinblat,
983 R. Oulton, M. Brongersma, S. A. Maier, [Non-plasmonic nanoantennas for](https://doi.org/10.1038/ncomms8915)
984 [surface enhanced spectroscopies with ultra-low heat conversion](https://doi.org/10.1038/ncomms8915), *Nature Com-*
985 *munications* 6 (2015) 7915. [doi:10.1038/ncomms8915](https://doi.org/10.1038/ncomms8915).
986 URL <https://doi.org/10.1038/ncomms8915>
- 987 [65] FAO, The state of food and agriculture 2022: Leveraging agricultural au-
988 tomation for transforming agrifood systems, Tech. rep., Food and Agriculture
989 Organization of the United Nations, Rome (2022).
- 990 [66] ICVCM, Assessment framework for high-integrity carbon credits, Tech. rep.,
991 Integrity Council for the Voluntary Carbon Market, version 2.0 (2024).